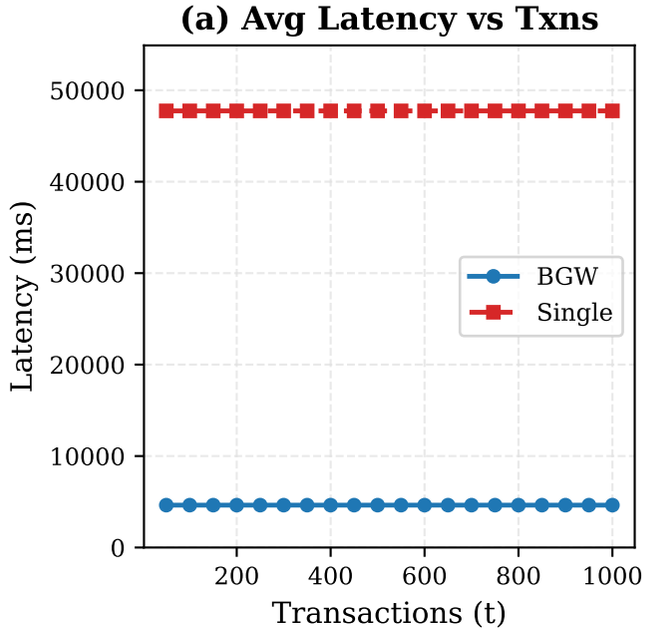
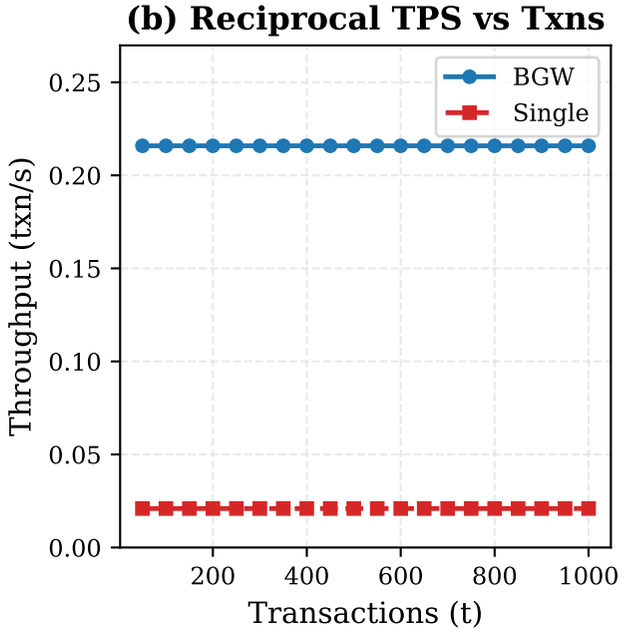
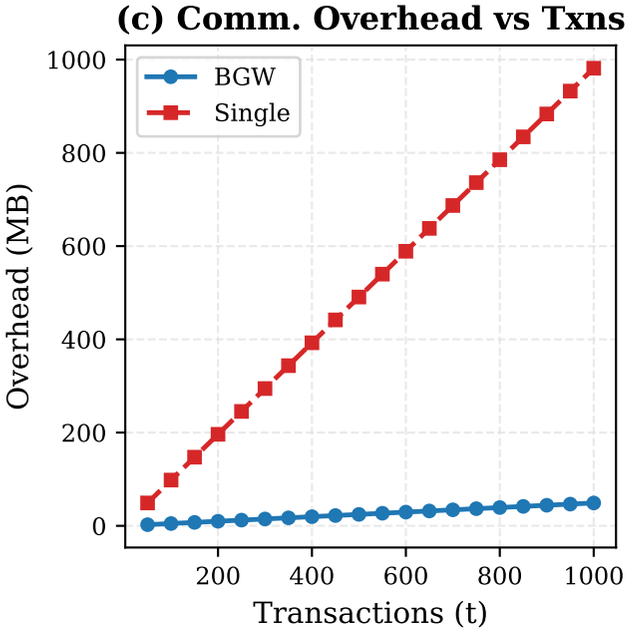
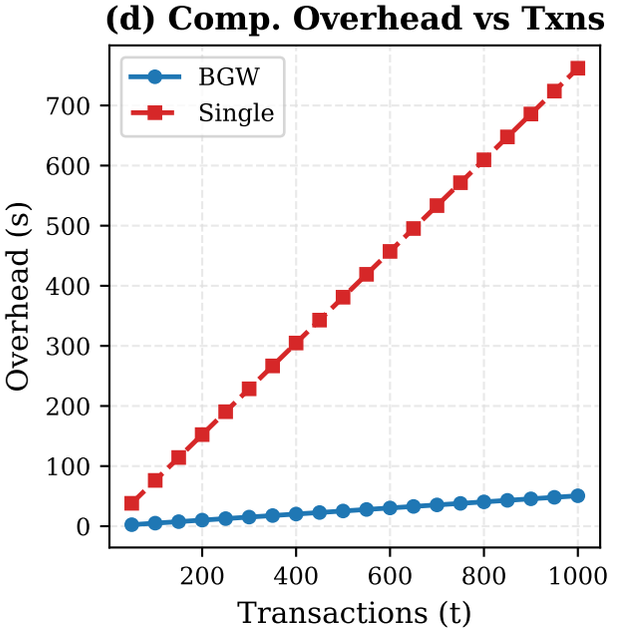
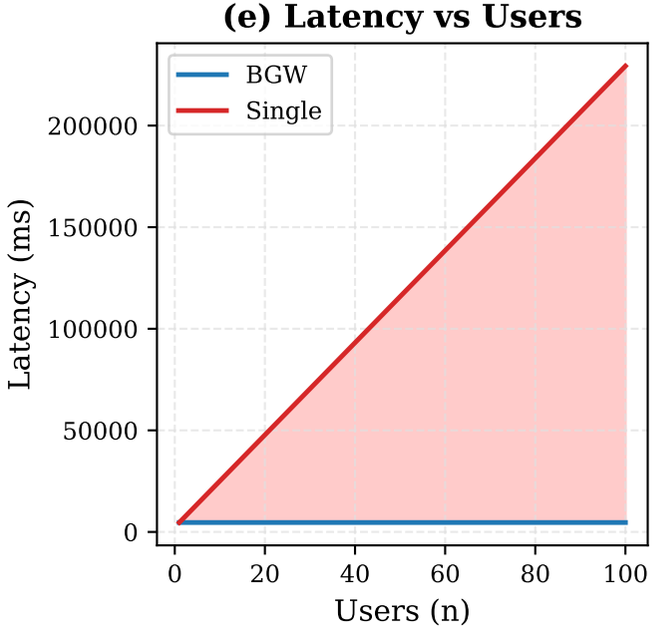
Analytical Complexity Model: BGW Broadcast Encryption vs Single Encryption
[Empirically Calibrated (Live System)]

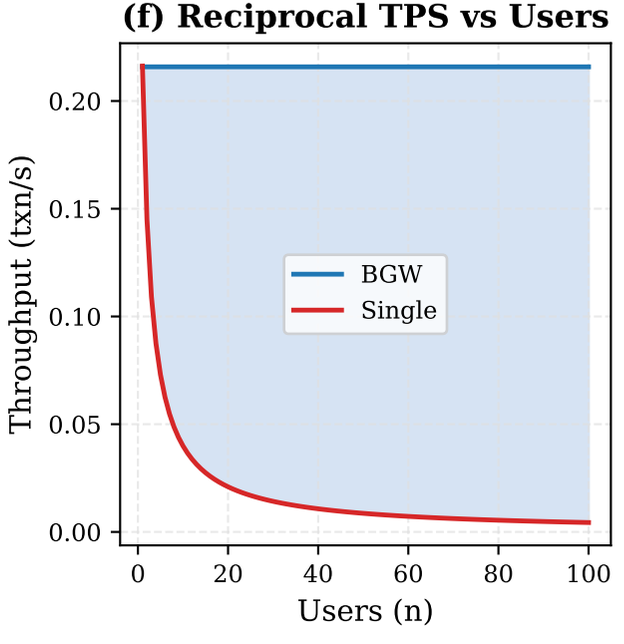


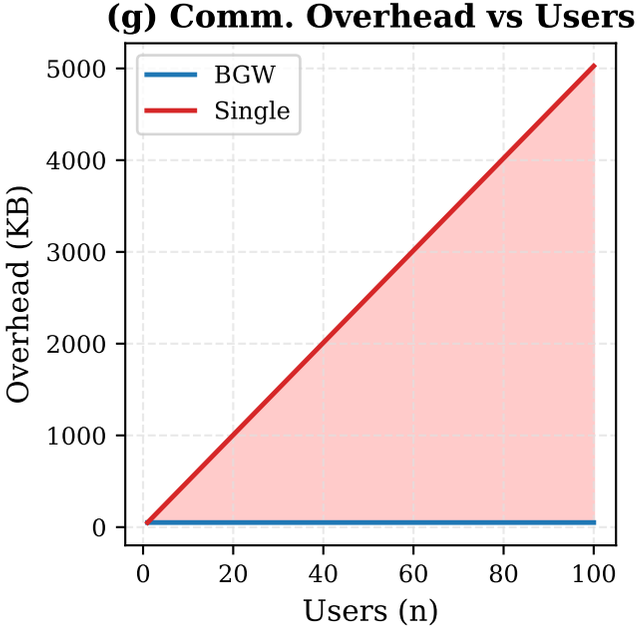












Empirically Calibrated (Live System)

```
T_AES        = 38.1 ms
T_BGW_hdr    = 12.5 ms
T_IPFS       = 2226.3 ms
T_BC         = 2355.7 ms
T_key_send   = 5.0 ms
Hdr Size     = 197 B (BGW)
Key Token    = 262 B (Single)
File Size D = 50.0 KB
n (for vs t)= 20

Model: Initial Multi-Share
BGW: 1 cipher + 1 header
Single: n encryptions + keys
```